

POOJA KIRAN BHARADWAJ

Security Engineering | Detection & Response | AI Security | Cloud Security
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Greater Phoenix Area, AZ | F-1 OPT EAD | Open to relocation

Security engineer focused on detection engineering, security telemetry, threat analysis, and automated response for cloud and AI infrastructure. Build security systems that normalize runtime events, identify attacker behavior, correlate multi-stage attack activity, enforce security decisions, and produce auditable evidence. Hands-on experience with Elastic SIEM, Kubernetes, AWS IAM, security automation, MITRE ATT&CK, adversarial testing, and AI security. Built and maintain open-source security systems spanning agent runtime security, cloud IAM attack-path analysis, model supply-chain security, and adversarial detection.

EXPERIENCE

Independent AI Security Researcher & Engineer

Self-Directed Security Research (concurrent with M.S. program) — Tempe, AZ · Aug. 2024 – Present

- Design and maintain 15 open-source security repositories spanning detection engineering, security enforcement, threat analysis, cloud IAM, AI agent security, model supply-chain security, and adversarial testing.
- Engineer detection pipelines that normalize security events, evaluate attacker behavior, correlate multi-stage activity, classify findings by severity, map findings to MITRE ATT&CK and MITRE ATLAS, and export SARIF for CI.
- Build automated security controls that make allow, block, redact, quarantine, and deny decisions with circuit breakers, rate limiting, tamper-evident audit logging, and explicit failure-mode handling.
- Validate security controls through automated unit, integration, failure-mode, and adversarial tests, including 569 tests for the MCP security gateway and positive/negative tests for all 25 AWS IAM rules plus failure-mode coverage.
- Integrate CodeQL, Bandit, Trivy, Grype, Syft SBOM, pip-audit, and multi-version CI to continuously validate security and software supply-chain controls.

Business Compliance Lead & Market/Cost Analyst

ASU Technology Innovation Lab (Honeywell Aerospace partnership) — Tempe, AZ · Aug. 2025 – Dec. 2025

- Evaluated cybersecurity product requirements against aerospace compliance frameworks and translated security risks into system-level architecture constraints.
- Conducted competitive and market analysis to evaluate security requirements, product capabilities, and implementation tradeoffs.

Graduate Teaching Assistant, IT Grader

Ira A. Fulton Schools of Engineering, Arizona State University — Mesa, AZ · Jan. 2025 – Oct. 2025

- Assessed cloud security architectures, network defense configurations, and IAM implementations for 100+ students per semester.
- Provided structured technical feedback on security architecture, implementation quality, threat exposure, and defensive controls.

SECURITY ENGINEERING PROJECTS

MCP Agent Security Gateway (17 stars) | Python, FastAPI, Docker, Kubernetes · Jan. 2025 – Aug. 2026

- Inline security proxy that intercepts MCP agent-to-tool calls via JSON-RPC stdio, evaluates each request through a 5-layer detection pipeline (server trust, tool-call policy, process-spawn analysis, semantic intent, network egress), and returns allow/block/redact/quarantine decisions before requests reach downstream servers.
- Detection engine: 50+ prompt-injection rules with Unicode normalization, zero-width stripping, homoglyph detection, Base64/ROT13 decode; PII detection across 9 data types; exfiltration-pattern matching for hidden recipients, oversized payloads, and raw-IP destinations.
- Operational controls: SHA-256 hash-chained audit logging, write-ahead log, circuit breakers (fail-closed), rate limiting, shadow mode, W3C traceparent propagation, Prometheus metrics; Kubernetes manifests with HPA and probes.
- Detection-engineering lab: exports decisions in Elastic Common Schema (ECS), ships to Elasticsearch/Kibana via Filebeat, runs an in-memory correlation engine (6 multi-event rules), 9 Elastic Security rules, and 6 Atomic Red Team style simulations, all mapped to MITRE ATT&CK.
- 569 tests, 75% coverage across Python 3.10-3.12. CI: CodeQL, Bandit, Trivy, Grype, Syft SBOM, pip-audit, Docker build.

AWS Agent Identity Guard | Python, AWS IAM, Terraform, SARIF · Mar. 2025 – Aug. 2026

- Static analyzer that detects dangerous IAM permission patterns in AI agent roles. 25 rules covering wildcard service grants, iam:PassRole without conditions, privilege-management actions, audit-trail tampering (CloudTrail, GuardDuty, Config), credential-harvest chains, lateral-movement combinations, and cross-account trust without ExternalId.
- Multi-step attack-path detection: identifies when a single identity combines credential harvesting, metadata reachability, and lateral-movement permissions into an exploitable chain that single-rule scanners miss.
- Outputs Text, JSON, and SARIF. Exit code 1 on critical/high findings serves as a CI merge gate; integrates with GitHub Code Scanning. Optional read-only live-account scan via boto3. Positive/negative tests for all 25 rules plus failure-mode coverage.

HF Model Provenance Scanner | Python, Hugging Face API, SARIF · Sep. 2024 – Aug. 2026

- Supply-chain scanner for Hugging Face model repositories. Analyzes pickle opcodes (Protocol 0-5), SafeTensors headers, GGUF structures, ONNX, and Keras files without downloading full weights; detects malicious deserialization, typosquatting, obfuscated code, and provenance gaps.
- Temporal baseline diffing for rug-pull detection, Levenshtein typosquat detection, multi-layer obfuscation decode, and CycloneDX 1.6 SBOM generation. Findings mapped to MITRE ATT&CK v19.

LLM Red-Team Framework | Python, scikit-learn, FastAPI, SARIF · Feb. 2025 – Jul. 2026

- Offline adversarial evaluation harness for prompt-injection detectors. Generates attack corpora across 6 categories mapped to OWASP LLM Top 10, trains a baseline TF-IDF classifier, and measures performance using grouped template splits that prevent data leakage. F1 = 0.97 on held-out templates; honestly reports degradation on novel phrasings.
- FastAPI service exposes /scan with rate limiting and API-key auth. SARIF output integrates with GitHub Code Scanning. 162 tests, 96% coverage.

EDUCATION

Arizona State University — Tempe, AZ

M.S. Information Technology · Aug. 2024 – May 2026

Relevant coursework: Machine Learning Security, Cloud Security Architecture, Information Assurance, Network Defense.

M. S. Ramaiah University of Applied Sciences — Bengaluru, India

B.Tech Computer Science & Engineering · Aug. 2019 – Aug. 2023

TECHNICAL SKILLS

Detection & Response: Detection engineering, SIEM integration, Elastic/ELK, ECS, event correlation, attack-chain detection, security telemetry, threat analysis, attack simulation, threat modeling, audit logging, policy enforcement, security automation, SARIF

Cloud & Infrastructure: AWS IAM, Kubernetes, Docker, Terraform, GitHub Actions, Elasticsearch, Kibana, Filebeat, Prometheus, Grafana

AI Security: AI agent runtime security, prompt-injection detection, model supply-chain security, LLM security evaluation, adversarial ML, data-poisoning detection, adversarial robustness

Threat Intel & Frameworks: MITRE ATT&CK v19, MITRE ATLAS, OWASP Top 10 for LLMs, NIST AI RMF, CycloneDX SBOM

Languages & Frameworks: Python, SQL, FastAPI, PyTorch, scikit-learn, NumPy

Security Tooling: CodeQL, Bandit, Trivy, Gype, pip-audit, Syft, Dependabot

PUBLICATION & RECOGNITION

- **IEEE INDICON 2023:** "A Personalized E-Learning System Using Reinforcement Learning Through Satellite" (IEEE Xplore: 10440852).
- **AWS Academy Graduate:** Cloud Security Foundations.
- **KSCST Research Grant:** Funded undergraduate research project.